Title

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.

I. INTRODUCTION

$8\pi G_{\rm N} = 1$?, spacetime signature $(+,-,-,-)$

II. SETUP

A. Lorentzian Regge calculus

Regge calculus [1] is a discrete gravitational theory on simplicial manifolds Δ with the length of edges $\{l_e\}$ dynamical. In a Lorentzian setting, a simplicial manifold consists of intrinsically flat Lorentzian 4-simplices $\sigma \in \Delta$. Curvature is located at triangles $t \in \Delta$ and captured by deficit angles $\delta_t(\{l_e\})$, lying in the space orthogonal to t. For t spacelike (timelike), this space is isomorphic to $\mathbb{R}^{1,1}$ ($\mathbb{R}^2$).[1] Lorentzian deficit angles are defined as $\delta_t^{\rm L,\pm} = \mp i 2\pi - \sum_{\sigma \supset t} \psi_{\sigma,t}^{\rm L,\pm}$, following the conventions of [2, 3], where the $\psi_{\sigma,t}$ are dihedral angles. "±" indicates a choice of sign which becomes important for causally irregular configurations [2–4] discussed below. The bulk action of Lorentzian Regge calculus is given by [1–3]

$$\begin{aligned} S_{\rm R}[\{l_e\}] = &\sum_{t\in\Delta:\ \rm tl} |A_t(\{l_e\})|\delta_t^{\rm E}(\{l_e\}) \\ &+ \sum_{t\in\Delta:\ \rm sl} |A_t(\{l_e\})|\delta_t^{\rm L,\pm}(\{l_e\})\,, \end{aligned} \tag{1}$$

with $|A_t|$ the absolute value of the triangle area. In the presence of a boundary, $\partial\Delta \neq \emptyset$, $S_{\rm R}$ attains boundary terms such that overall additivity is ensured. The equations of motion are given by $\frac{\partial S_{\rm R}}{\partial l_e} = 0$ supplemented by the Schläfli identity, $\sum_t |A_t| \frac{\partial \delta_t}{\partial l_e} = 0$.

Regge calculus can be formulated in terms of different variables [5, 6], with area variables [7–9] playing a particularly important role here due to their connection to LQG and spin-foams. The relation between area and length variables is in general not invertible, and area variables need to be further constrained to yield the dynamical equations of length Regge calculus [3, 6, 9]. However, in the symmetry reduced setting introduced below, the relation of area and length is globally invertible and the aforementioned constraints trivialize. For the remainder, we study the classical dynamics of area Regge calculus which will be used in future work as a comparison to results of the effective spin-foam path integral.

B. Lorentzian 4-frusta

Lorentzian 4-dimensional frusta are ideal to capture spatially flat, homogeneous and isotropic discrete geometries [10, 11]. The boundary of a 4-frustum, depicted in Fig. 1, contains two spacelike 3-cubes and six 3-frusta that are either spacelike or timelike. Gluing 4-frusta along boundary 3-frusta in spatial direction and along boundary 3-cubes in temporal direction, one obtains an extended cosmological spacetime of hypercubical combinatorics $\mathcal{X}_{\mathcal{V}}^{(4)}$ with slices labelled by $n \in \{0, ..., \mathcal{V}_T\}$ discretized by $\mathcal{V}_S^3$ spacelike 3-cubes. We say that a 4-frustum lies in a "slab" between slices n and $n+1$.

The entire geometry of a slab is captured by three variables describing the 3-cubes and 3-frusta which can be either areas or lengths. The former set of variables consists of the area of squares contained in the 3-cubes, j_0 and j_1, and the area of the trapezoids contained in the 3-frusta, k. The latter consists of the edge length contained in the 3-cubes, s_0 and s_1, and the height of the 4-frustum, H_0, which implicitly determines the trapezoids and 3-frusta.[2] In the highly symmetry restricted setting considered here, there is a one-to-one map between area variables, (j_0, j_1, k_0), and length variables, (s_0, s_1, H_0), given explicitly in Appendix **??**. We will exploit this relation in Sec. IV using length variables for analytical computations and clear interpretation, especially because the height is directly related to the lapse function N in the continuum. When solving the Regge equations numerically thereafter, we will use area variables instead to provide a classical benchmark for the effective spin-foam

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[1] For the remainder, we neglect lightlike edges, triangles and tetrahedra. See [2] for the definition of deficit angles associated to $(d-1)$-dimensional null cells.

[2] Strictly speaking, the height H_0 is a derived quantity that is not a length associated to a 1-dimensional sub-cell of the cellular complex. Such a quantity would instead be given by the length l_0 associated to the strut connecting two neighboring spacelike slices. However, the relation between strut length and height is invertible for non-null struts, which assume here.

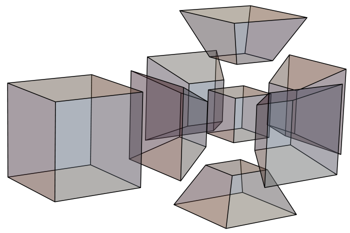

FIG. 1: Boundary of a 4-frustum given by two 3-cubes and six equal 3-frusta.

calculations of [?]. The general notions of Regge calculus introduced above are straightforwardly adapted to either of the two sets of variables.

The squared volume of subcells connecting neighboring slices, i.e. 3-frusta, trapezoids and edges, which are detailed in Appendix ??, carries information on the causal structure. If it is positive (negative), the corresponding subcell is timelike (spacelike). In the symmetry reduced setting, the ratio $\frac{|j_n - j_{n+1}|}{4k_n}$ fully characterizes the causal character of subcells suggesting separating the 4-frusta configurations into three sectors summarized in Tab. ??.

As discussed in [11–15], two of these sectors I and II exhibit an irregular light cone structure, also referred to as causality violations, located at spacelike sub-cells. Examples of such irregularities are given by the trouser singularity and the yarmulke singularity associated to spatial topology change [16–18]. For causality violations at $(d-2)$-dimensional spacelike hinges the imaginary parts of the dihedral angles do not sum up to $\mp i2\pi$, leading to a complex valued Regge action. Choosing "+" in the convention of Lorentzian angles, the amplitude $e^{iS_{\mathrm{R}}}$ is exponentially suppressed for trouser and exponentially enhanced for yarmulke configurations, vice versa for "−" [3].Without a complexification of the variables, the Regge equations of motion for causality violating configurations are complex equations with real variables and therefore overconstrained. Moreover, the results of [11, 12] show that in the respective settings, no real solutions exist. Since this applies also to the setting used in this work we henceforth focus on the causally regular sector III.

C. Lorentzian Regge action for spatially flat cosmology

For a single 4-frustum between, say slice n and $n+1$, the causally regular Regge action is given by

$$S_{\mathrm{III}}^{(n)} = 6(j_n - j_{n+1}) \sinh^{-1}\left(\frac{j_{n+1} - j_n}{\sqrt{16k_n^2 + (j_n - j_{n+1})^2}}\right) + 12k_n \left[\frac{\pi}{2} - \cos^{-1}\left(\frac{(j_n - j_{n+1})^2}{16k_n^2 + (j_n - j_{n+1})^2}\right)\right] - \Lambda V^{(4)}\,, \tag{2}$$

where Λ is a cosmological constant and $V^{(4)}$ is the 4-volume defined in Eq. (??) as an equation of (j_n, j_{n+1}, k_n). By construction, the 4-frustum action is additive such that on a discretization $\mathcal{X}_{\mathcal{V}}^{(4)}$ with $\mathcal{V}_S^3$ vertices in spatial direction and $\mathcal{V}_T + 1$ spatial slices, the action is given by $S_{\mathrm{R}} = \mathcal{V}_S^3 \sum_{n=0}^{\mathcal{V}_T - 1} S_{\mathrm{R}}^{(n)}$. Additivity of the action per slab implies that the equations of motion for a given geometric quantity, being either j_n or k_n, will only depend on neighboring geometric labels.

In the following, we study the discrete dynamics of spatially flat cosmology as governed by the Regge equations. To that end, we consider first the case of a single time step, $\mathcal{V}_T = 1$, in Sec. III and generalize then to an arbitrary number of time steps in Sec. IV.

III. RECAP: ONE TIME STEP

The classical and quantum dynamics of a single timestep in discrete Lorentzian cosmology have been studied extensively in [4, 12, 13] for spatial curvature with cosmological constant and in [11, 15] for spatially flat spacetimes with a free scalar field. In this chapter we extend these studies to spatially flat cosmology with a cosmological constant, before then focussing on the case of arbitrarily many time steps.

For $\mathcal{V}_T = 1$, the square areas j_0, j_1 are boundary data and only the trapezoid area k_0 is dynamical. Thus, the Regge equations are simply given by

$$\frac{\partial S_{\mathrm{III}}}{\partial k_0} = 0\,, \tag{3}$$

where due to the Schläfli identity, the first term in Eq. (2) drops out. Notice that this equation is transcendental and can therefore only by tackled numerically.

Vanishing cosmological constant. In the case of $\Lambda = 0$, the Regge action describes a spatially flat vacuum universe. If the boundary data is non-stationary, i.e. $j_0 \neq j_1$, Eq. (3) exhibits no solution for k_0. In other words, a classical universe can neither expand nor contract if there is no matter present. This is in accordance with the continuum Friedmann equations in vacuum, $\left(\frac{\dot{a}}{a}\right)^2 = 0$, with a the scale factor, which are solved only by $a = $ const. On the other hand, if $j_0 = j_1$, Eq. (3) is trivially satisfied and the trapezoid area k_0 is undetermined. In fact, the Regge action vanishes, $S_{\mathrm{III}} = 0$ for all k_0, and the 4-frusta reduce to 4-cuboids discretizing flat space. The arbitrariness of k is a common phenomenon for flat solutions in Regge calculus [19, 20] and signifies the restoration of diffeomorphism symmetry in the discrete [21, 22]. This symmetry amounts to moving spacelike slices (instead of individual vertices), see also the discussion in [23]. Note that for 4-cuboids, $k = \sqrt{j}H$ with H the height of the cuboid which is directly related to the lapse function N in the continuum [11, 12]. The lapse is the gauge variable of the residual diffeomorphism symmetry after symmetry reduction, encoding time reparametrization invariance.

Non-vanishing cosmological constant. For a non-vanishing cosmological constant, $\Lambda \neq 0$, and stationary boundary conditions, $j_0 = j_1$, the only solution to Eq. (3) is $k_0 = 0$, corresponding to degenerate 4-frusta with vanishing height. Thus, a stationary universe for $\Lambda \neq 0$ only exists if no time passes, i.e. if there is no actual time evolution. The more interesting case is $j_0 \neq j_1$ where a non-trivial solution k_0^* for Eq. (3) exists. More precisely, a unique solution k_0^* is found for any $\Lambda > 0$ and any $j_0 \neq j_1$. The on-shell trapezoid area $k_0^*(\Lambda)$ as a function of Λ behaves as a monomial decay $k_0^* \sim c\Lambda^{-\alpha}$ with $c, \alpha > 0$ constants depending on j_0, j_1. For $\Lambda \neq 0$, the solution diverges and ceases to exist at $\Lambda = 0$ in lines with the discussion above. In the limit $\Lambda \to \infty$, Eq. (3) is dominated by the Λ-term, i.e.

$$\Lambda \frac{(j_0 + j_1)^2 k}{4V^{(4)}} \approx 0 \,, \tag{4}$$

solved by $k_0^* = 0$.

Note that the restriction on the sign of Λ is in accordance with the continuum Friedmann equations, $\left(\frac{\dot{a}}{a}\right)^2 = N^2 \frac{\Lambda}{3}$, where real solutions for the scale factor require $\Lambda > 0$.

For completeness, let us contrast these results with the existence of solutions in the spatially spherical case in Lorentzian [12] and Euclidean [24, 25] signature.[3] In the former setting, note first that for $\Lambda = 0$, no solutions exist. Furthermore, for $\Lambda > 0$, solutions exist only if $j_0 > j_\Lambda$ and $j_1 > j_\Lambda$, where j_Λ is a Λ-dependent minimal area. This is in accordance with the continuum where the scale factor typically follows a cosh-function, attaining a minimal value a_Λ. In Euclidean signature, solutions exist for $\Lambda = 0$. These correspond, in the continuum, to solutions where $\dot{a} = \text{const}$. For $\Lambda > 0$, solutions exist only if $j_0 < j_\Lambda$ and $j_1 < j_\Lambda$ where j_Λ is the same as in the Lorentzian case, however now representing an upper bound. Again, this behavior agrees with continuum results where a Euclidean universe with positive spatial curvature can only reach a maximum value a_Λ.

[3] The construction of the spatially spherical Regge action differs from the construction here, in particular because there exist only finitely many homogeneous triangulations of the 3-sphere, see [12, 25] for details.

IV. ARBITRARY NUMBER OF TIME STEPS

In this section, we study the Regge equations of spatially flat Lorentzian cosmology for an arbitrary number of time steps $\mathcal{V}_T$. This amounts to solving $2\mathcal{V}_T - 1$ coupled transcendental equations for the bulk variables $k_0, j_1, \dots, j_{\mathcal{V}_T - 1}, k_{\mathcal{V}_T - 1}$ for given boundary data $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ,

$$\frac{\partial S_{\text{III}}}{\partial k_n} = 0 \,, \qquad \frac{\partial S_{\text{III}}}{\partial j_m} = 0 \,, \tag{5}$$

with $n \in \{0, \dots, \mathcal{V}_T - 1\}$ and $m \in \{1, \dots, \mathcal{V}_T - 1\}$. Since the Regge action S_{III} is additive with respect to the slices, i.e. $S_{\text{III}} = \mathcal{V}_S^3 \sum_n S_{\text{III}}^{(n)}$, Eqs. (5) simplify significantly,

$$\begin{aligned} 0 &= \frac{\partial S_{\text{III}}}{\partial k_n} = \mathcal{V}_S^3 \frac{\partial S_{\text{III}}^{(n)}}{\partial k_n} \,, \\ 0 &= \frac{\partial S_{\text{III}}}{\partial j_n} = \mathcal{V}_S^3 \left(\frac{\partial S_{\text{III}}^{(n)}}{\partial j_n} + \frac{\partial S_{\text{III}}^{(n+1)}}{\partial j_n} \right) . \end{aligned} \tag{6}$$

Thus, it is sufficient to consider the single slab action $S_{\text{III}}^{(n)}$, given in Eq. (3), and its derivatives. An explicit form of these equations is given in Appendix **??**.

A priori, the number of cubes in each spacelike slice, $\mathcal{V}_S^3$, enters the total Regge action only as a multiplicative factor due to the symmetry restriction. Thus, in the following two sections, we consider $\mathcal{V}_S = 1$ for simplicity. However, let us mention here that in Sec. IV C, the parameter $\mathcal{V}_S$ turns out to be crucial to obtain the correct continuum limit. We tackle the problem of finding solutions to Eq. (5) first analytically in the subsequent section and develop numerical methods thereafter.

Add explicit equations to the appendix, mention simple dependence, additivity

A. Analytical considerations

For vanishing cosmological constant, $\Lambda = 0$, and stationary boundary data, $j_0 = j_{\mathcal{V}_T}$, the results for a single time step obtained in the previous section straightforwardly generalize. That is, if $\Lambda = 0$, the Regge equations are satisfied if and only if the boundary data satisfies $j_0 = j_{\mathcal{V}_T}$. Then, the bulk areas $j_n = j_0$ for all $n \in \{1, \dots, \mathcal{V}_T - 1\}$, and the trapezoid areas are undetermined, marking the restoration of time reparametrization invariance of flat space. On the other hand, for $\Lambda \neq 0$ and boundary data $j_0 = j_{\mathcal{V}_T}$, no non-degenerate solution to the Regge equations is found. A unique solution exist for $j_0 \neq j_{\mathcal{V}_T}$ when $\Lambda > 0$ which is in accordance with the continuum Friedmann equations. To summarize, a spatially flat vacuum universe is flat and stationary while a positive cosmological will necessarily lead to curved non-stationary solutions. For the remainder, we assume $\Lambda > 0$ and $j_0 \neq j_{\mathcal{V}_T}$.

Eq. (5) has been analyzed further (coupling a scalar field rather than a cosmological constant) in [11].

Therein, the length variables (s_n, s_{n+1}, H_n) have been employed in which the monotonicity of solutions and the continuum limit are derived most easily. The results of [11] transferred to the present setting with $\Lambda \neq 0$ are briefly summarized in the following.

Combining the Regge equations

$$\frac{\partial S_{\mathrm{III}}}{\partial H_{n-1}} = 0\,, \quad \frac{\partial S_{\mathrm{III}}}{\partial H_n} = 0\,, \quad \frac{\partial S_{\mathrm{III}}}{\partial s_n} = 0\,, \tag{7}$$

one can straightforwardly derive the relation [11]

$$\frac{s_n - s_{n-1}}{H_{n-1}} = \frac{s_{n+1} - s_n}{H_n}\,, \tag{8}$$

for all $n \in \{1, \ldots, \mathcal{V}_T - 1\}$. That is, the difference of spatial edge length with respect to the height of the 4-frusta remains constant. Introducing

$$\tau_0 = \sum_{n=0}^{\mathcal{V}_T - 1} H_n\,, \tag{9}$$

which we will later identify as the discrete elapsed proper time, one can show that Eq. (8) implies

$$\frac{s_{n+1} - s_n}{H_n} = \frac{s_{\mathcal{V}_T} - s_0}{\tau_0}\,, \tag{10}$$

for all n. Since $H_n, \tau_0 > 0$, this implies $\mathrm{sgn}(s_{n+1} - s_n) = \mathrm{sgn}(s_{\mathcal{V}_T} - s_0)$, that is, solutions evolve monotonically and the boundary data $s_0, s_{\mathcal{V}_T}$ selects whether the universe is expanding, $s_{\mathcal{V}_T} > s_0$, or contracting, $s_{\mathcal{V}_T} < s_0$.

Following the procedure outlined in [11], an analytical continuum limit of the Regge equations can be taken in two steps. First, one performs a linearization of $\frac{\partial S_{\mathrm{III}}}{\partial H_n} = 0$ in the variable $\eta_n = \frac{s_{n+1} - s_n}{H_n}$. In this approximation, corresponding to small deficit angles, one finds

$$\frac{(s_n - s_{n+1})^2}{s_n^2 + s_{n+1}^2} = H_n^2 \frac{\Lambda}{6}\,. \tag{11}$$

Then, the discrete variables (s_n, s_{n+1}, H_n) need to be identified with the continuum scale factor $a(t)$ and the lapse function $N(t)$ parametrized in some coordinate time t. In the continuum limit, the height H_n corresponds to the infinitesimal proper time, $H_n \to N\mathrm{d}t$. The discrete length on a given slice is mapped to the scale factor, $s_n \to a(t)$, while the length of a neighboring edge attains an additional time derivative term, $s_{n+1} \to a(t) + \dot{a}\mathrm{d}t$. Then, at leading order in $\mathrm{d}t$, on finds indeed that Eq. (11) approaches

$$\left(\frac{\dot{a}}{a}\right)^2 = N^2 \frac{\Lambda}{3}\,, \tag{12}$$

which is the Friedmann equation for a spatially flat universe with cosmological constant. Note that configurations with spacelike trapezoids or spacelike 3-frusta satisfy $\eta_n^2 > 2$ and therefore bear no straightforward continuum limit [11].

The cosmological Regge equations have been shown to capture the Friedmann equations in a suitable continuum limit in previous works [10–12, 26]. Still, Eq. (5) remain to be solved explicitly in the discrete for many time steps $\mathcal{V}_T$. This represents however an important step since the solutions serve as a classical and discrete benchmark for identifying a semi-classical continuum regime in discrete path integral approaches such as effective spin-foams. In the following section, we tackle this issue employing numerical methods.

B. Numerical implementation

Solving the Regge equations explicitly requires numerical methods. In this work, we employ the package `NLsolve`[4] in `Julia`, designed for solving non-linear equations. Note that while Eqs. (5) define a real-valued function in real variables, `NLsolve` can also be applied to complex numbers which may be relevant for the complex critical point method []. The algorithm we developed for this work specifically tailored to Eq. (5) amounts to the `solve_eom`[5] method which we explain in the following.

First, note that the method `nsolve` of the `NSolve` package takes as input a vector valued function $\vec{F}$, a vector of initial variables $\vec{x}_0$, and parameters for the number of iterations as well as the error tolerance for solutions. If the problem is sufficiently well-behaved, `nsolve` finds a solution to the equation $\vec{F}(\vec{x}) = \vec{0}$ using $\vec{x}_0$ as a starting guess of the solving algorithm. In our case, $\vec{F} : \mathbb{R}^{2\mathcal{V}_T - 1} \longrightarrow \mathbb{R}^{2\mathcal{V}_T - 1}$, with

$$\vec{F} = \left(\frac{\partial S_{\mathrm{III}}}{\partial k_0}, \frac{\partial S_{\mathrm{III}}}{\partial j_1}, \ldots, \frac{\partial S_{\mathrm{III}}}{\partial j_{\mathcal{V}_T - 1}}, \frac{\partial S_{\mathrm{III}}}{\partial k_{\mathcal{V}_T - 1}}\right)\,, \tag{13}$$

$$\vec{x} = (k_0, j_1, \ldots, j_{\mathcal{V}_T - 1}, k_{\mathcal{V}_T - 1})\,. \tag{14}$$

Therefore, adapting Eq. (5) to be solved by `nsolve` is done in two steps: 1) set up the function $\vec{F}$ for fixed boundary areas $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ, and 2) propose a starting guess $\vec{x}_0$. While step 1) is straightforward, step 2) requires particular care because the starting guess $\vec{x}_0$ is crucial for convergence and good performance of the solving algorithm. In particular, as $\mathcal{V}_T$ grows, the number of variables and equation grows and the algorithm because more sensitive to a well-chosen $\vec{x}_0$.

By trial and error, one can find sufficient starting guesses for `nsolve` to converge given various small $\mathcal{V}_T$.[6]

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[4] A documentation of the package can be found at github.com/JuliaNLSolvers/NLsolve.jl. Note that also the `NSolve` method in `Mathematica` can be used, but is expected have lower performance in terms of accuracy and computation time.

[5] See this github repository for the source code.

[6] We tried using the continuum solution to find a good initial guess. However, we found that the prescription given here ensures fast convergence more reliably.

For the solutions $\vec{x}^* = (k_0^*, j_1^*, \ldots, j_{\mathcal{V}_T-1}^*, k_{\mathcal{V}_T-1}^*)$, the following patterns are being observed. First, the j_n^* can be approximated as

$$j_n^* \approx j_0 + n\Delta j\,, \quad n \in \{1, \ldots, \mathcal{V}_T - 1\}\,, \tag{15}$$

with $\Delta j = (j_{\mathcal{V}_T} - j_0)/\mathcal{V}_T$. Second, let $\boldsymbol{k}_0^*$ be the solution of the trapezoid area for $\mathcal{V}_T = 1$. Then, for $\mathcal{V}_T > 1$, the k_n^* are approximated by

$$k_n^* \approx \boldsymbol{k}_0^*/\mathcal{V}_T\,, \quad n \in \{0, \ldots, \mathcal{V}_T - 1\}\,. \tag{16}$$

That is, to find a good initial guess for the trapezoid areas, one only needs to find a solution $\boldsymbol{k}_0^*$ for a single time step. Crucially, for a single equation, `nsolve` converges for a broad range of initial guesses, so $\boldsymbol{k}_0^*$ can be easily found. In summary, finding a good initial guess $\vec{x}_0$ amounts to finding the one time step solution $\boldsymbol{k}_0^*$ and then setting

$$\vec{x}_0 = \left(\frac{\boldsymbol{k}_0^*}{\mathcal{V}_T}, j_0 + \Delta j, \ldots, j_0 + (\mathcal{V}_T - 1)\Delta j, \frac{\boldsymbol{k}_0^*}{\mathcal{V}_T}\right)\,. \tag{17}$$

solver summary

To understand the stability of the solver under perturbations of the initial guess $\vec{x}_0$, consider a vector $\vec{\delta} = \frac{1}{\sqrt{2\mathcal{V}_T - 1}}(\delta, \ldots, \delta)$. Applying the solver with $\vec{x}_0 + \vec{\delta}$ as initial guess one finds the following behavior:

- For $\mathcal{V}_T = 1$, the solver yields consistent results as long as the entries of $\vec{x}_0 + \vec{\delta}$ are positive. For negative entries, there are non-trivial sign changes in the Regge equation, and in this case, it not clear if solutions even exist. Choosing δ such that the entries are positive, the solutions are stable even for large $\delta > 100$. Also, the number of iterations does only increase slightly. This is important since setting up $\vec{x}_0$ requires finding $\boldsymbol{k}_0^*$ first. If this first step would be extremely sensitive, the proposal above would not be viable.

- For $\mathcal{V}_T > 1$, the number of iterations needed to find convergence is generically larger, even for $\delta = 0$. Choosing δ such that the entries of $\vec{x}_0 + \vec{\delta}$ are positive, one finds that solutions are stable for small $|\delta|$ ($|\delta| \lesssim 10$ for $\mathcal{V}_T = 2$) with little fluctuations in the required iterations. However, for larger δ, the solutions suddenly jump to unreasonably large or to negative values and the iterations also increase slightly. This effect increases with $\mathcal{V}_T$. For the solver, this means that 1) one should stick with the proposal of $\vec{x}_0$ given above, and 2) for very large $\mathcal{V}_T$, one needs to come up with a refined proposal of initial guesses as $\vec{x}_0$. In this work, the values we consider for $\mathcal{V}_T$ are small enough for $\vec{x}_0$ to yield satisfactory results.

, one observes that for

- output
- convergence monitor
- scaling of computation time as function of $\mathcal{V}$

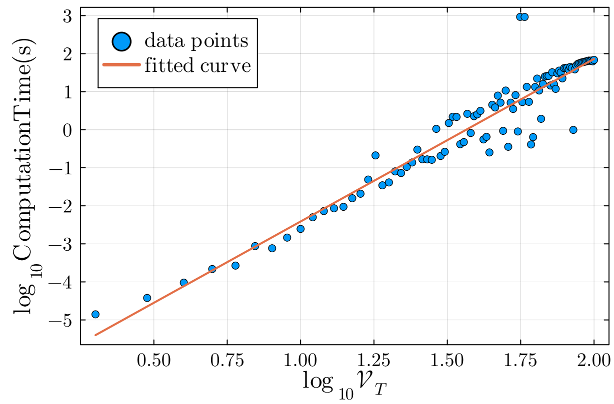

FIG. 2: Fit with $\hat{\beta} = 4.23$ and $\hat{\alpha} = 6.67$. $t(V) = 10^{\alpha} V^{\beta}$.

C. Continuum limit

Facilitated by the numerical tools developed above, we study in this section the continuum limit of Regge calculus at the level of the solutions, rather than at the level of the equations. Quantifying whether the discrete solutions provide a satisfactory approximation to the continuum solutions requires two ingredients: 1) For comparison, one needs to consider appropriate continuum quantities which serve as limiting points of the discrete solutions. 2) One needs to identify the right parameters to tune. Here, the discreteness is parametrized by the number of slices, $\mathcal{V}_T$, as well as the number of 3-cubes per slice, $\mathcal{V}_S^3$. Thus, the continuum is expected to be reached by suitably scaling $\mathcal{V}_T, \mathcal{V}_S \to \infty$.

1) Continuum quantities. One difficulty in comparing discrete and continuum solutions is that in the discrete, the trapezoid areas are dynamical while in the continuum, the lapse function N is a gauge variable, and thus, any time parametrization corresponds to a gauge choice. That is, when performing the limit, which time parametrization in the continuum emerges? Clearly, the labels $n \in \{0, \ldots, \mathcal{V}_T\}$ do not serve as a time as this would imply constant distance between any two neighboring spacelike slices. Instead, recall that the height of 4-frusta was mapped to infinitesimal proper time in the analytical continuum limit of Sec. **??**. Therefore, the sum of heights, $\mathcal{T} = \sum_n H_n$, serves as a discrete approximation of the continuum proper time elapsed, defined as $\int \mathrm{d}t\, N(t)$. The advantage of considering proper time is that it is gauge invariant. Thus, one does not run into the issue of having to pick a parametrization in the continuum to compare to the discrete solutions. Using proper time, we will then compare the evolution of the square area j_n^* as a function of the accumulated height $\sum_{m=0}^{n} H_m$ with the squared scale factor $A = a^2$ as a function of proper time.

2) The right scaling. A priori, the parameter $\mathcal{V}_S$ enters the Regge action only as a multiplicative factor. Therefore, it might be tempting to simply set $\mathcal{V}_S = 1$, as done in the analytical part in Sec. **??**. However, Fig. **??**

demonstrates that leaving $\mathcal{V}_S = 1$ while increasing $\mathcal{V}_T$, the accumulated height $\tau = \sum H$ does not converge to the value of the continuum proper time τ_0. To obtain the correct τ_0, one therefore needs to tune $\mathcal{V}_S$ and $\mathcal{V}_T$ simultaneously. Here, a second subtlety needs to be noticed. The continuum proper time only depends on Λ and the initial and final squared scale factor A_0, A_1. Setting $j_0 = A_0$ and $j_{\mathcal{V}_T} = A_1$ and increasing $\mathcal{V}_S$ i

$\mathcal{V}_T$ and $\mathcal{V}_S$ can be scaled to larger values individually. In particular, from Eq. (5), one can see that

Continuum limit also in the analytical sense? Consider $s_n \to s_n/\mathcal{V}_S$.

V. CONCLUSION

Extension to $k = \pm 1$.

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Appendix A: Frusta geometry

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